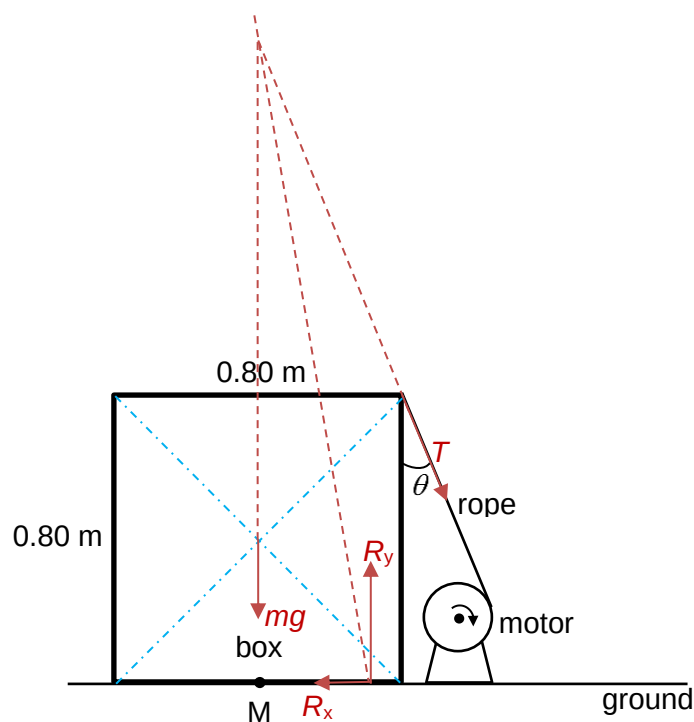




- (a) Since the wheel is in rotational equilibrium,

$$Tr = \tau$$

$$T = \frac{\tau}{r} = \frac{5.0}{0.20} = 25 \text{ N}$$

- (b) Since the box is in equilibrium,  
horizontally, (not required to answer this part)

$$T_x - R_x = 0$$

$$R_x = T \sin 20^\circ$$

vertically,

$$R_y - T_y - mg = 0$$

$$R_y = T \cos 20^\circ + mg$$

By the principle of moments, taking moments about point M,

$$R_y d = T_x h + T_y \left( \frac{h}{2} \right)$$

$$d = \frac{h}{R_y} \left( T_x + \frac{1}{2} T_y \right)$$

$$= \frac{0.80}{T \cos 20^\circ + mg} \left( T \sin 20^\circ + \frac{T \cos 20^\circ}{2} \right)$$

$$= \frac{0.80T}{T \cos 20^\circ + mg} (\sin 20^\circ + 0.5 \cos 20^\circ)$$

$$= \frac{0.80(25)}{25 \cos 20^\circ + 2.0(9.81)} (\sin 20^\circ + 0.5 \cos 20^\circ)$$

$$= 0.3766 = 0.377 \text{ m}$$

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**(c)** Solution 1

As  $\theta$  increases, the increased clockwise moment about point M due to the tension exceeds the maximum anticlockwise moment possible due to the normal contact force when the normal contact force is at the corner ( $d = 0.40 \text{ m}$ ) of the box. The box will rotate clockwise about the bottom right corner.

Solution 2

As  $\theta$  increases,  $d$  increases to balance the increasing moment due to the tension and if  $d > 0.40 \text{ m}$ , this implies that the contact force needs to act outside the box for the box to be in equilibrium. Since it is not possible for the contact force to act outside the box, there is a maximum value for  $\theta$ .

Solution 3

As  $\theta$  increases, the horizontal component of the tension in the rope increases. If the horizontal component exceeds the maximum friction on the box by the ground, the box will slide.